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Our application aims to help users evaluate and compare neighborhoods across the United States to determine where it may be best to live. By integrating multiple datasets—including housing prices, crime statistics, public school information, and IRS income data—we link these datasets using ZIP codes to provide a comprehensive view of each area.
The platform will allow users to explore key indicators such as average housing prices, local crime rates, nearby school availability, and the income levels of residents in each ZIP code. By combining these factors, the application enables users to make more informed decisions about potential places to live based on safety, affordability, education, and socioeconomic conditions.

Dataset 1: USA Real Estate Dataset - Link
The USA Real Estate dataset contains approximately 2.23 million records with more than 10 attributes, including price, number of bedrooms, bathrooms, and location. The price attribute has an average value of approximately $524,000 with a standard deviation of about $2.14 million, indicating large variation in housing prices. The bed attribute has an average of 3.28 bedrooms, and the bath attribute has an average of 2.5 bathrooms per property. The state attribute includes 55 unique states, with Florida accounting for 11% of listings.

Dataset 2: USA Crime Dataset - Link
The US Crime dataset contains approximately 638,000 records with about 10 attributes, including agency information, city, state, and crime type. The Agency Type attribute shows that Municipal Police account for 77% of crime reports and Sheriff departments account for 16%. The City attribute includes 1,782 unique cities, with Los Angeles (7%) and New York (6%) being the most common. The State attribute shows that California accounts for 16% of reported crimes and Texas accounts for 10%, indicating that large states contribute a significant portion of the dataset.

Dataset 3: Public School Name and Address Dataset - Link
This dataset contains 35,291 schools in the U.S. for the 2009 school year. The numeric attribute member09 (student enrollment) has an average of 527 students per school, a standard deviation of 502, and ranges from -2 to 8,077, showing substantial variation in school sizes. Categorical variables indicate that most schools are level 1, followed by levels 3, 2, 4, and N. In terms of school type (type09), the majority are type 1 (30,967 schools), with smaller numbers in types 2–5. Most schools have status09 = 1, representing active schools, while other codes correspond to different administrative statuses. Overall, these statistics provide a clear overview of school sizes, types, levels, and geographic distribution.

Dataset 4: SOI Tax Stats - Individual income tax statistics - 2022 ZIP Code data (SOI) - Link

This dataset contains aggregated IRS individual income tax statistics for Tax Year 2022, organized by ZIP code. It includes approximately 67,204 records, covering 11,158 ZIP codes across 23 states. Key attributes include N1 (number of tax returns), A00100 (Adjusted Gross Income), A00200 (wages), and A00300 (interest income), with monetary values reported in thousands of dollars. The average number of tax returns per record is about 2,198, and the mean Adjusted Gross Income is approximately 213,638 (thousand dollars) with a standard deviation of 5,660,150, indicating large variation in income across ZIP codes.

SQL List:

1. Find the top 10 cities and states with the highest average house prices, considering only listings where the price is available. Only include cities that have at least 50 listings, and sort the results by the average price in descending order.
2. Use city and state as foreign keys to JOIN Dataset 1: USA Real Estate and Dataset and Dataset 2: USA Crime Dataset to get the amount of crime event of different city.
3. Use mzip as foreign key to JOIN Dataset 3: Public School and 1: USA Real Estate and Dataset and get the amount of different level public school of different zip code.
4. To quantify the association between housing prices and crime intensity on an annual basis, we will perform an inner join between Dataset 2 (USA Crime Dataset) and Dataset 1 (USA Real Estate Dataset). Next, we will group the joined records like city, state, year), calculate the annual total incident count, and compute price summaries for the housing market.
5. We will link Dataset 1: USA Real Estate and Dataset, Dataset 2: USA Crime Dataset and Dataset 3: Public School by the same city and year to build a unified location–year table. For each location and year, we will aggregate total crime incidents and attach the corresponding education metrics and housing price level.